

Username:

CS441

2025-09-24

Name:

Discrete Structures for CS

hw_4

Problem 1. Show that the additive inverse, or negative, of an even number is an even number using a direct proof.

Answer.

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Problem 2. Prove that if n is a perfect square, then $n + 2$ is not a perfect square.

Answer.

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Problem 3. Prove that if x , y , and z are integers and $x + y + z$ is odd, then at least one of x , y , and z is odd.

Answer.

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Problem 4. Prove that $m^2 = n^2$ if and only if $m = n$ or $m = \neg n$.

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Problem 5. Show that the product of two of the numbers $65^{1000} - 8^{2001} + 3^{177}$, $79^{1212} - 9^{2399} + 2^{2001}$, and $24^{4493} - 5^{8192} + 7^{1777}$ is nonnegative. Is your proof constructive or non-constructive? [Hint: Do not try to evaluate these numbers!]

Answer.

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Problem 6. Show that if r is an irrational number, there is a unique integer n such that the distance between r and n is less than $\frac{1}{2}$.

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Problem 7. The **quadratic mean** of two real numbers x and y equals $\sqrt{\frac{x^2+y^2}{2}}$. By computing the arithmetic and quadratic means of different pairs of positive real numbers, formulate a conjecture about their relative sizes and prove your conjecture.

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Problem 8. Prove that there are no solutions in integers x and y to the equation $2x^2 + 5y^2 = 14$.

Answer.

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Challenge 1. Prove that $\sqrt[3]{2}$ is irrational.

Answer.